

Fertilization

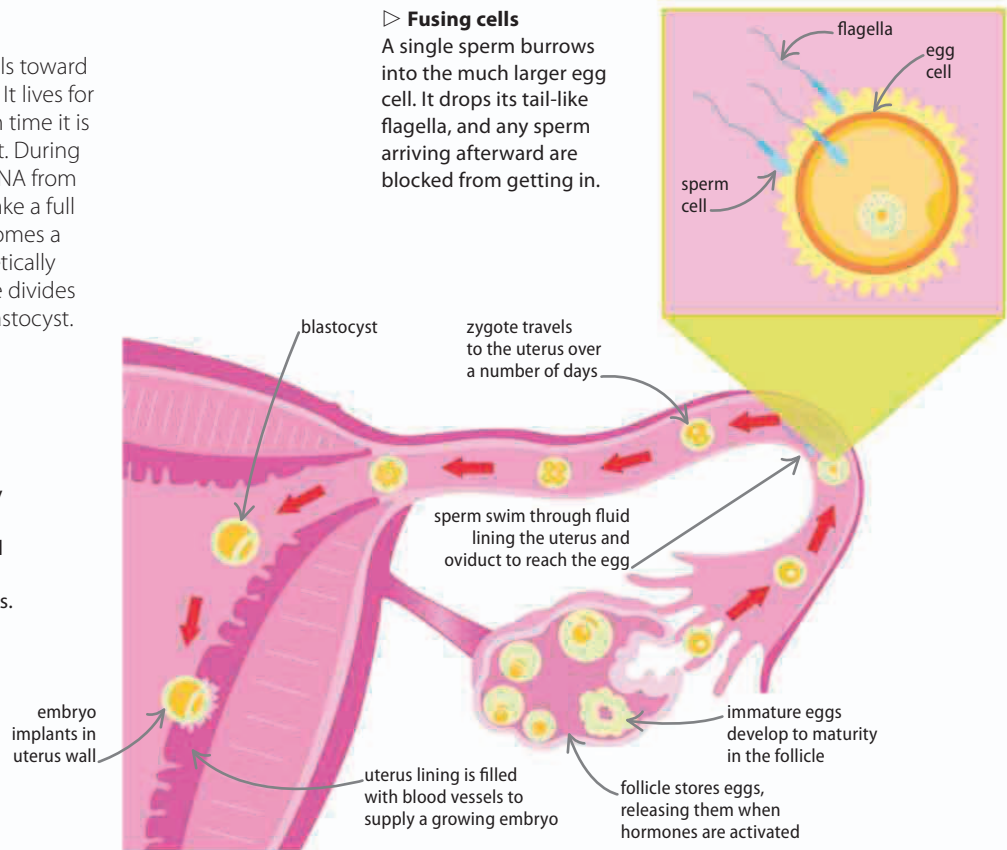
After ovulation, the egg travels toward the uterus along the oviduct. It lives for about 18 hours, during which time it is ready for a sperm to fertilize it. During fertilization, the half sets of DNA from both sex cells combine to make a full set. At this point the cell becomes a zygote, the first cell of a genetically unique individual. The zygote divides into a ball of cells, called a blastocyst.

► Implantation

The blastocyst can survive only for a few days on its own. It must implant in the uterus wall within about ten days in order to receive oxygen and nutrients. Once it does this, it continues to divide into new cells, and is then called an embryo.

► Fusing cells

A single sperm burrows into the much larger egg cell. It drops its tail-like flagella, and any sperm arriving afterward are blocked from getting in.

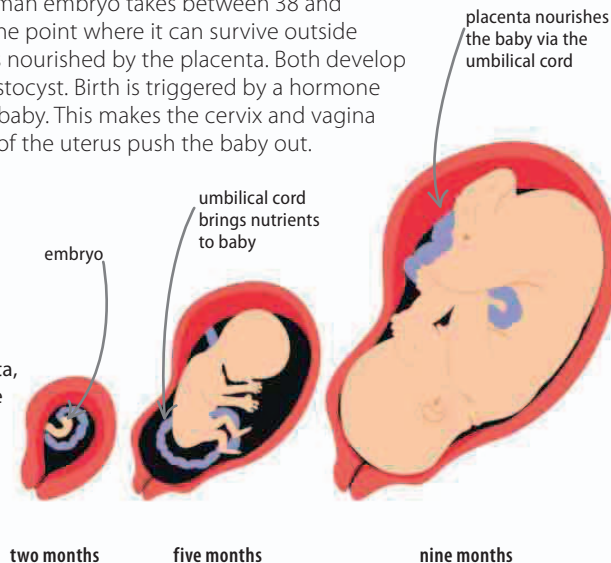


Gestation

From fertilization, the human embryo takes between 38 and 42 weeks to develop to the point where it can survive outside the uterus. The embryo is nourished by the placenta. Both develop from the same single blastocyst. Birth is triggered by a hormone released by the growing baby. This makes the cervix and vagina soften, and contractions of the uterus push the baby out.

► Early development

Every baby grows from a single cell called a zygote. This divides rapidly into a growing ball of cells. The cells differentiate into those that form the placenta, the membranes around the embryo, and the embryo itself. The cells that form the baby are identifiable after eight days of growth.



REAL WORLD

Fetal development

After about eight weeks of growth, the baby has all of its primary organs and recognizable human features. From this point it is known as a fetus. The development of a fetus can be monitored by scanning the womb with ultrasound to produce an image (below).

